

Visualizing the Geometry of Optimization: A Tutorial

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May 18, 2026

Outline:

- Linear programming
- Quadratic programming
- Geometry of quadratic function
- Mixed-integer quadratic programming

Linear Programming

Optimization problem:

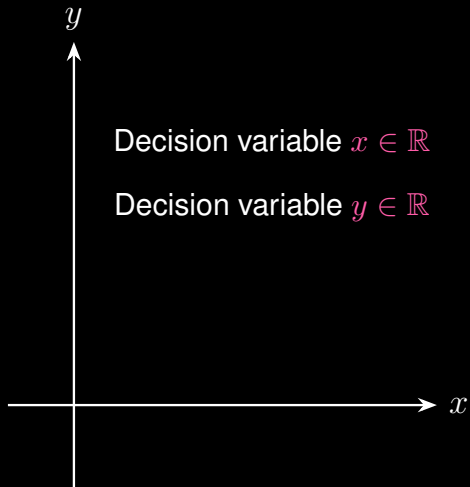
$$\max_{x \in \mathbb{R}, y \in \mathbb{R}} \quad 4x + y$$

$$\text{s.t.} \quad x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

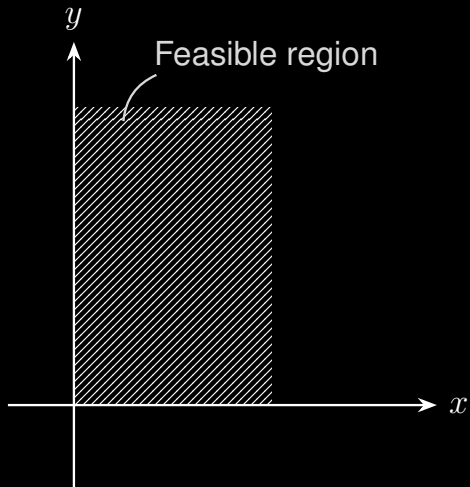
$$x \leq 3$$



Linear Programming

Optimization problem:

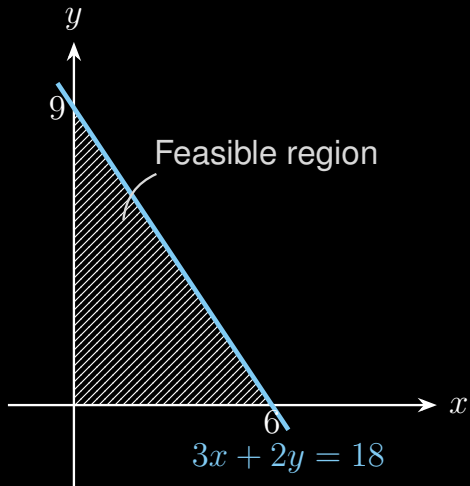
$$\begin{array}{ll}\max_{x \in \mathbb{R}, y \in \mathbb{R}} & 4x + y \\ \text{s.t.} & x, y \geq 0\end{array}$$



Linear Programming

Optimization problem:

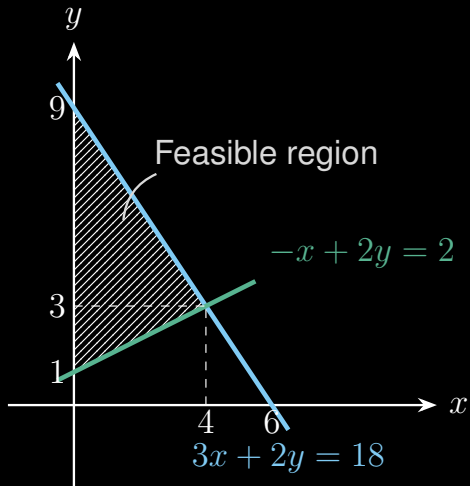
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Linear Programming

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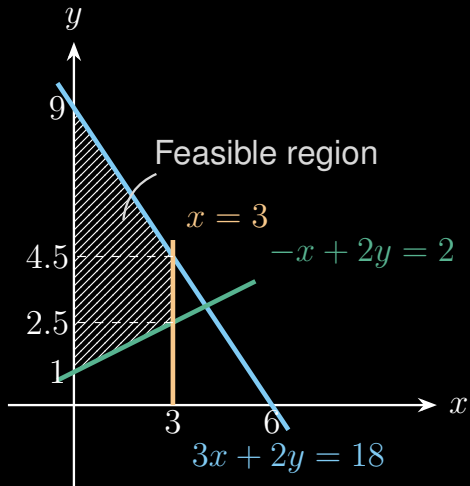
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Linear Programming

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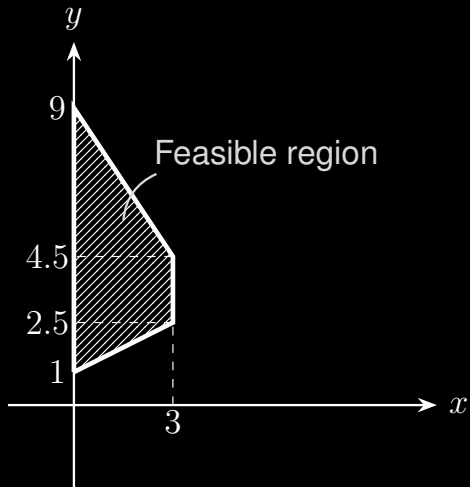
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Linear Programming

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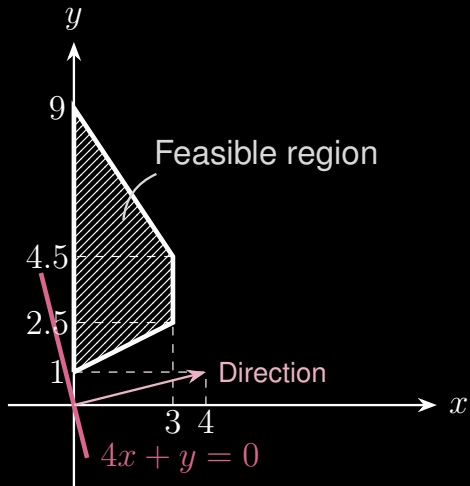


Linear Programming

Optimization problem:

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Direction vector $(4, 1)$

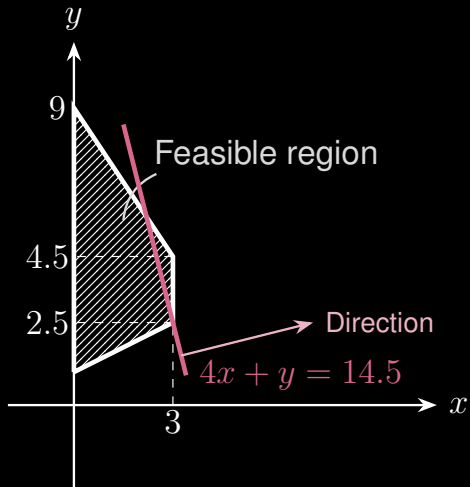


Linear Programming

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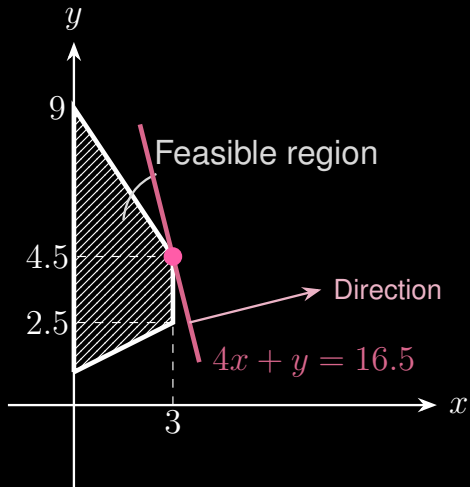


Linear Programming

Optimization problem:

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Optimal solution: $x = 3, y = 4.5$



How about integer decision variables?

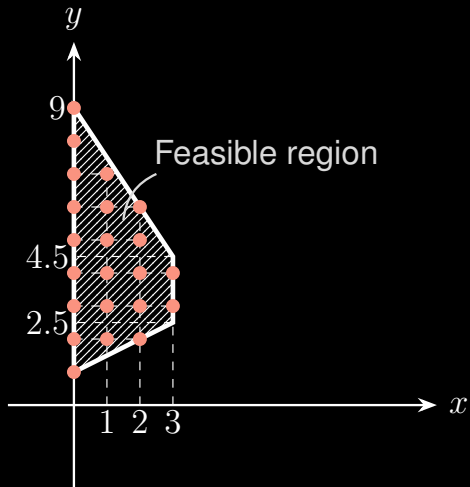
$$x \in \mathbb{Z} \text{ and } y \in \mathbb{Z}$$

Integer Linear Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \mathbb{Z}, y \in \mathbb{Z}} & 4x + y \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Feasible solutions (●)

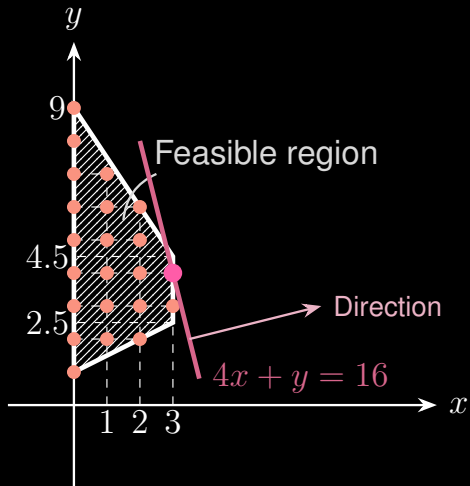


Integer Linear Programming

Optimization problem:

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Optimal solution: $x = 3, y = 4$



How about integer decision variables?

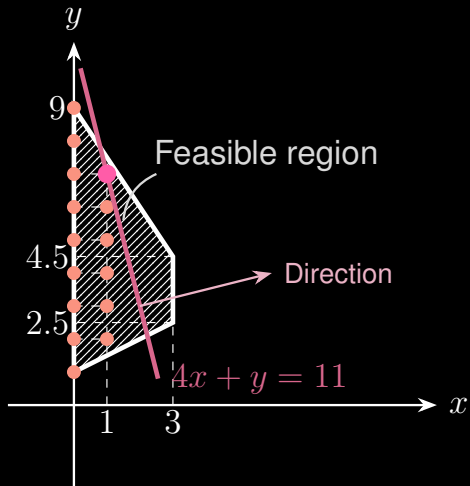
$$x \in \{0, 1\} \text{ and } y \in \mathbb{Z}$$

Integer Linear Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \{0,1\}, y \in \mathbb{Z}} & 4x + y \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Feasible solutions (● $x = 0, 1$)

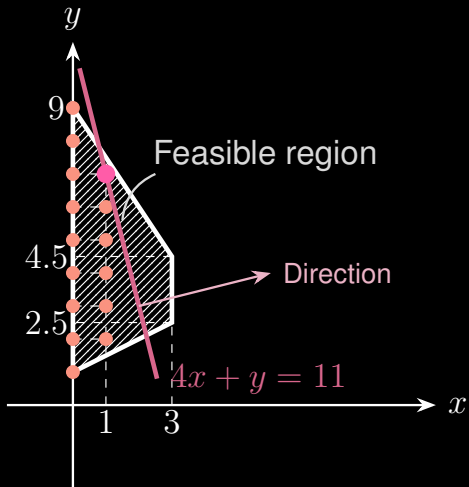


Integer Linear Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \{0,1\}, y \in \mathbb{Z}} & 4x + y \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Optimal solution: $x = 1, y = 7$



How about mixed-integer decision variables?

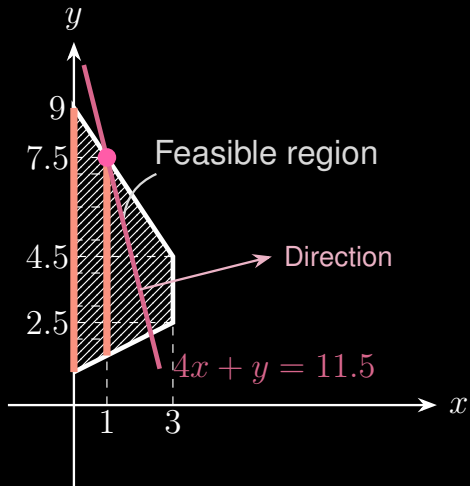
$$x \in \{0, 1\} \text{ and } y \in \mathbb{R}$$

Mixed-Integer Linear Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \{0,1\}, y \in \mathbb{R}} & 4x + y \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Feasible solutions (—)

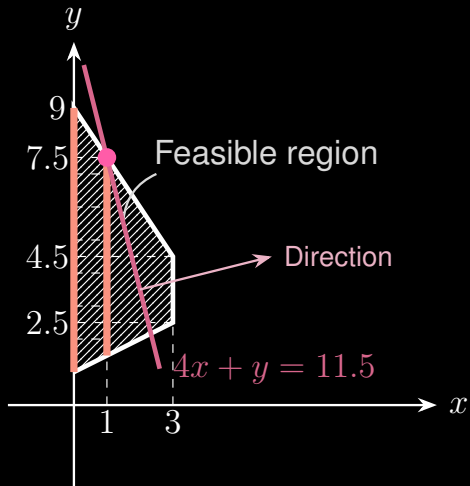


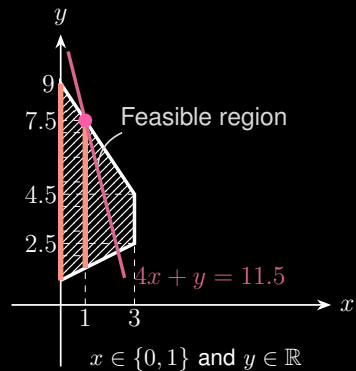
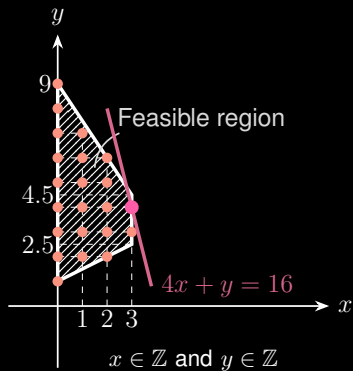
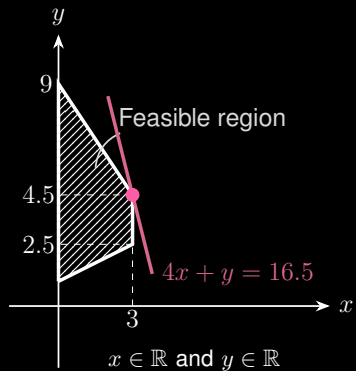
Mixed-Integer Linear Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \{0,1\}, y \in \mathbb{R}} & 4x + y \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Optimal solution: $x = 1, y = 7.5$





Outline:

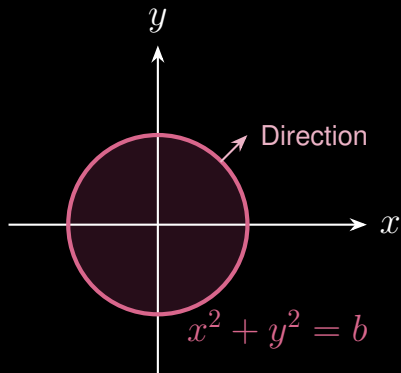
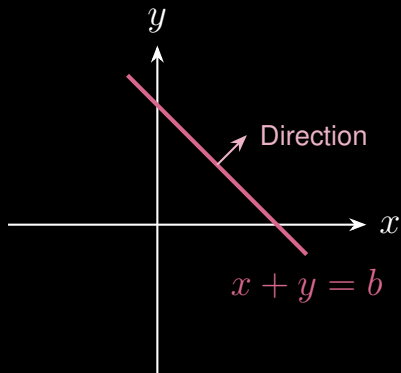
- ✓ Linear programming
- ✗ Quadratic programming
- ✗ Geometry of quadratic function
- ✗ Mixed-integer quadratic programming

How about quadratic objective function?

$$\max x + y$$

→

$$\max x^2 + y^2$$



Quadratic Programming

Optimization problem:

$$\max_{x \in \mathbb{R}, y \in \mathbb{R}} x^2 + y^2$$

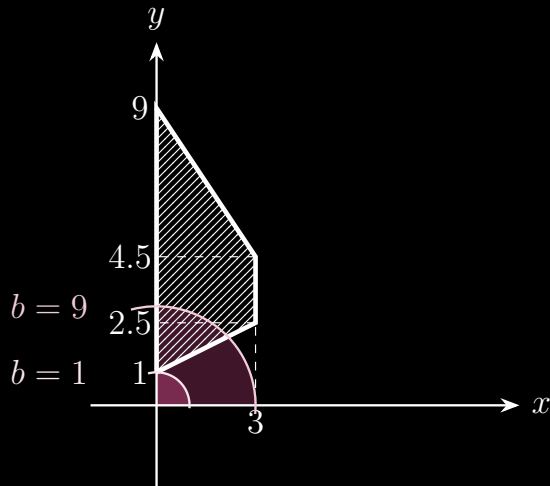
$$\text{s.t. } x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective function $x^2 + y^2 = b$



Quadratic Programming

Optimization problem:

$$\max_{x \in \mathbb{R}, y \in \mathbb{R}} x^2 + y^2$$

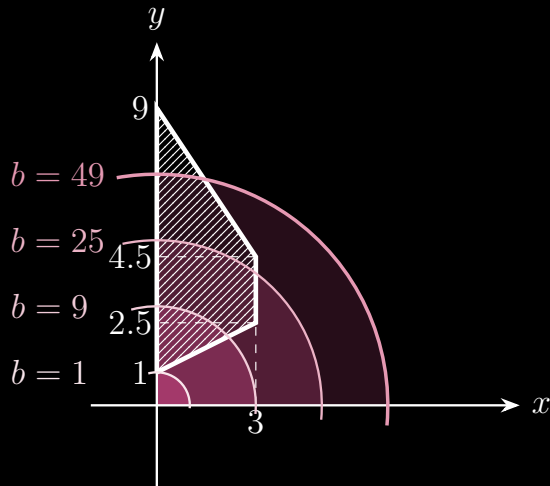
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$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective function $x^2 + y^2 = b$

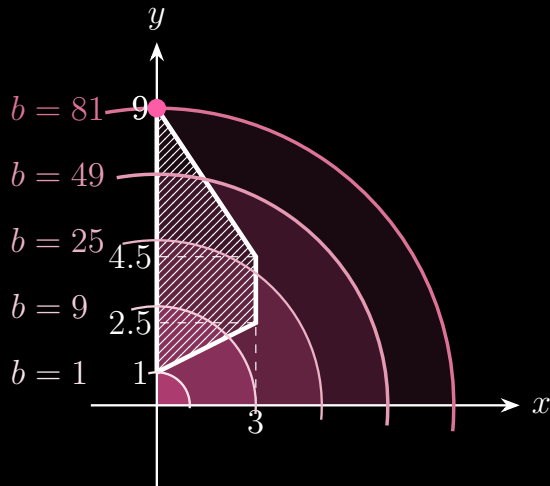


Quadratic Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \mathbb{R}, y \in \mathbb{R}} & x^2 + y^2 \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

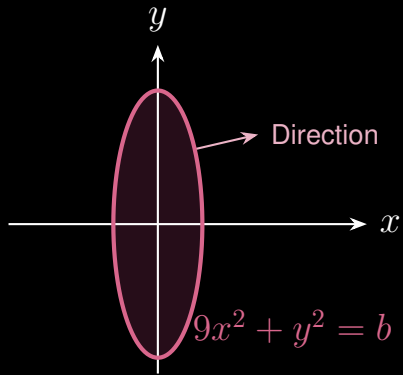
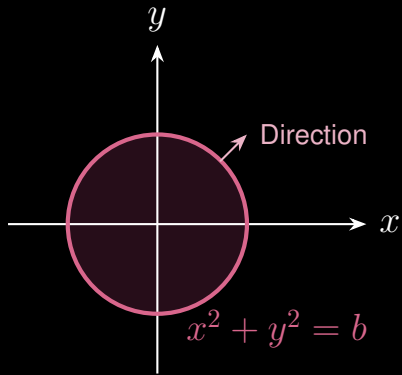
Optimal solution: $x = 0, y = 9$



$$\max x^2 + y^2$$

→

$$\max 9x^2 + y^2$$



Quadratic Programming

Optimization problem:

$$\max_{x \in \mathbb{R}, y \in \mathbb{R}} \quad 9x^2 + y^2$$

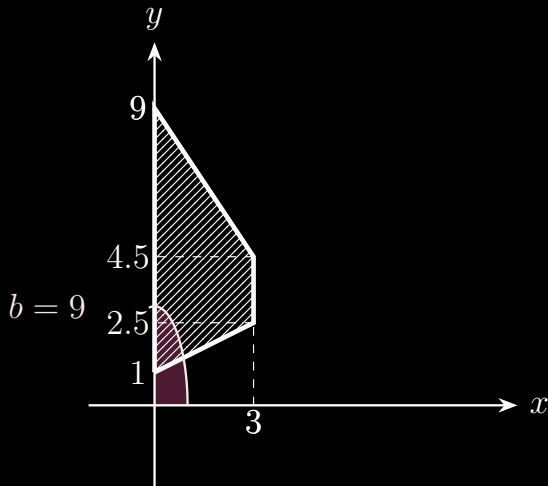
$$\text{s.t.} \quad x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective function $9x^2 + y^2 = b$



Quadratic Programming

Optimization problem:

$$\max_{x \in \mathbb{R}, y \in \mathbb{R}} \quad 9x^2 + y^2$$

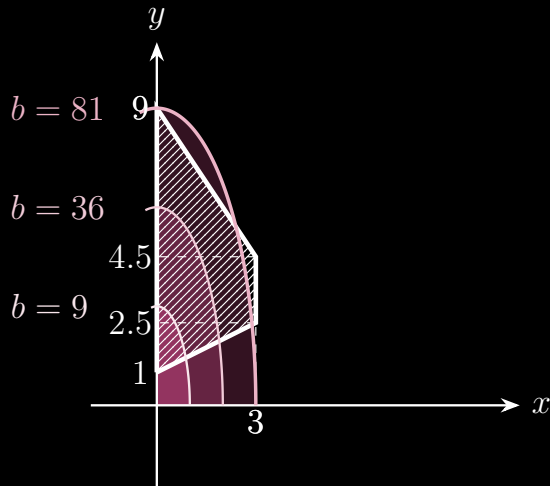
$$\text{s.t.} \quad x, y \geq 0$$

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$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective function $9x^2 + y^2 = b$

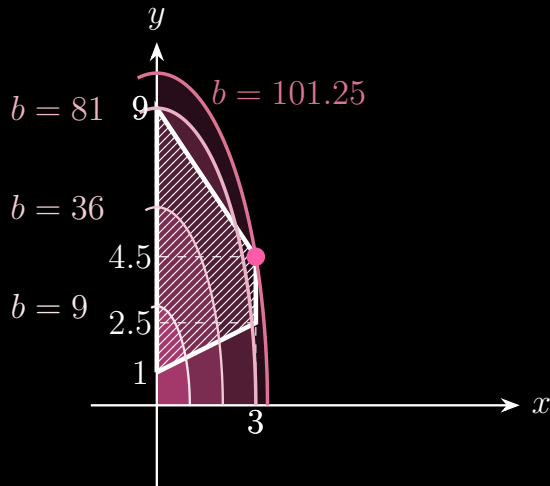


Quadratic Programming

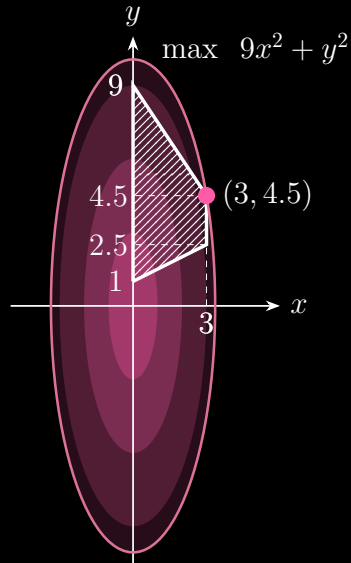
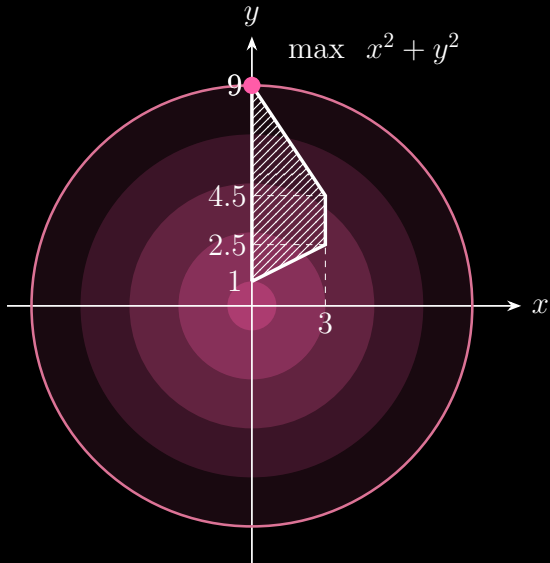
Optimization problem:

$$\begin{aligned} \max_{x \in \mathbb{R}, y \in \mathbb{R}} \quad & 9x^2 + y^2 \\ \text{s.t.} \quad & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3 \end{aligned}$$

Optimal solution: $x = 3, y = 4.5$



Different objectives $\rightarrow$ different optimal solutions

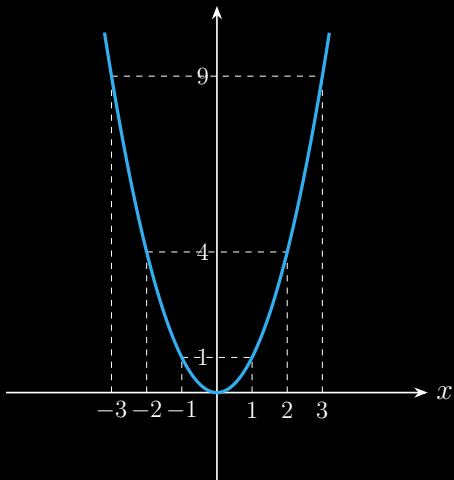


Outline:

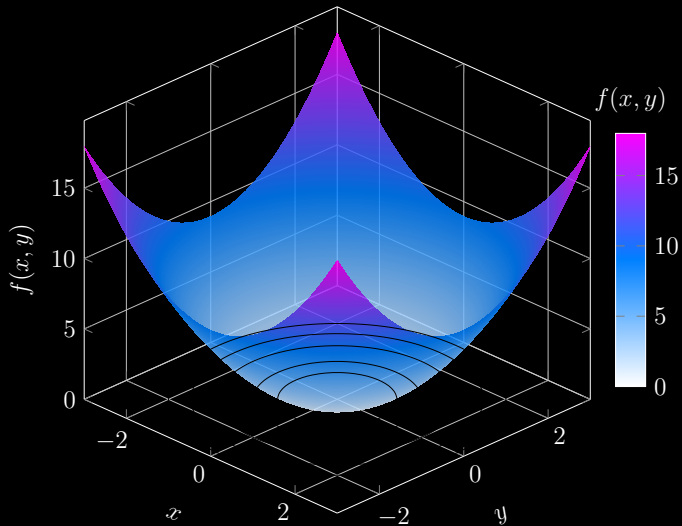
- ✓ Linear programming
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Quadratic Function

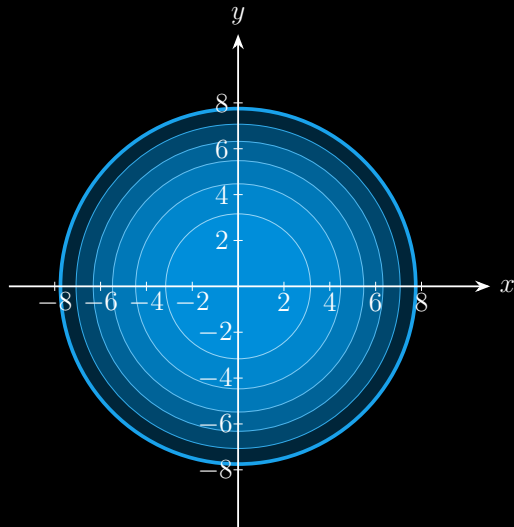
$$f(x) = x^2$$



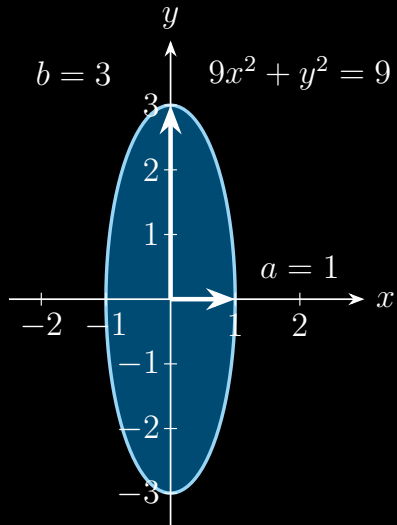
Quadratic Function $f(x, y) = x^2 + y^2$



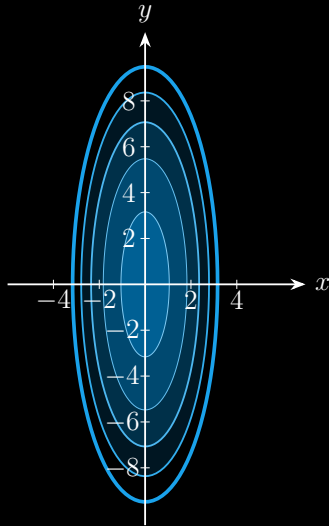
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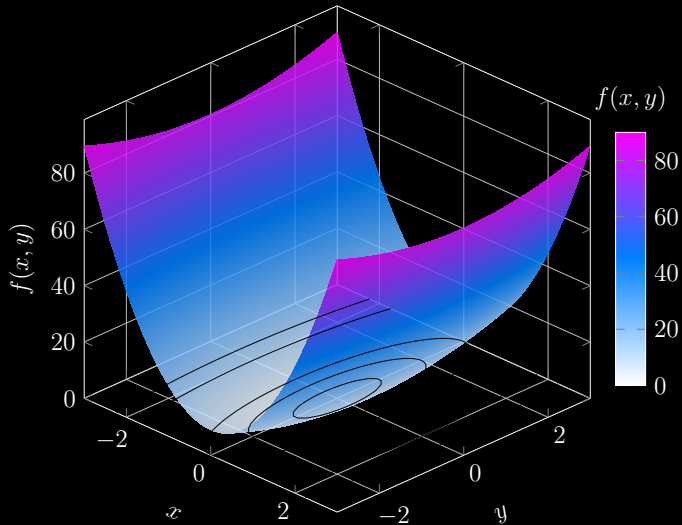
Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function $f(x, y) = 9x^2 + y^2$



Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = \begin{bmatrix} x \\ y \end{bmatrix}^T \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x \\ y \end{bmatrix}$$

Eigenvalue decomposition:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

```
1 import numpy as np
2
3 A = np.array([[9, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```


Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}^\top}_{\mathbf{w}^\top} \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{w}}$$

Using the eigenvalue decomposition:

$$\begin{aligned} \mathbf{w}^\top \mathbf{A} \mathbf{w} &= \mathbf{w}^\top (\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top) \mathbf{w} \\ &= \lambda_1 (\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2 (\mathbf{u}_2^\top \mathbf{w})^2 \end{aligned}$$

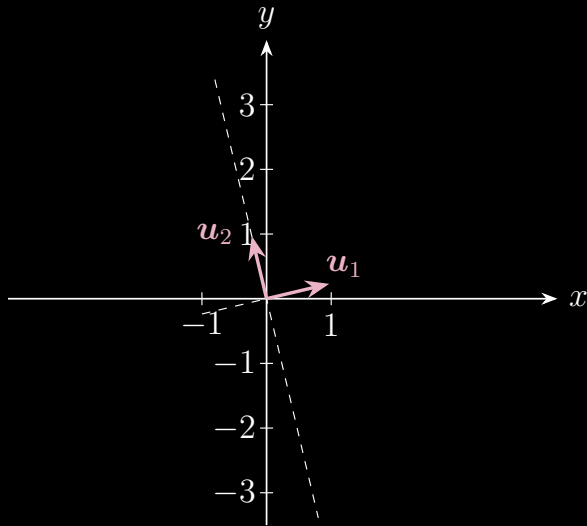
with eigenvalues and eigenvectors:

$$\lambda_1 = 9.47 \quad \mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \lambda_2 = 0.53 \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

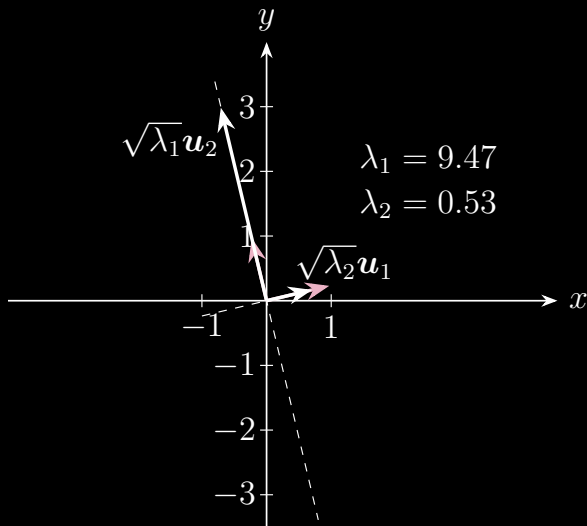
If $f(x, y) = \lambda_1 \lambda_2$, then the ellipse is

$$\lambda_1(\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2(\mathbf{u}_2^\top \mathbf{w})^2 = \lambda_1 \lambda_2 \quad \Rightarrow \quad \frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$$

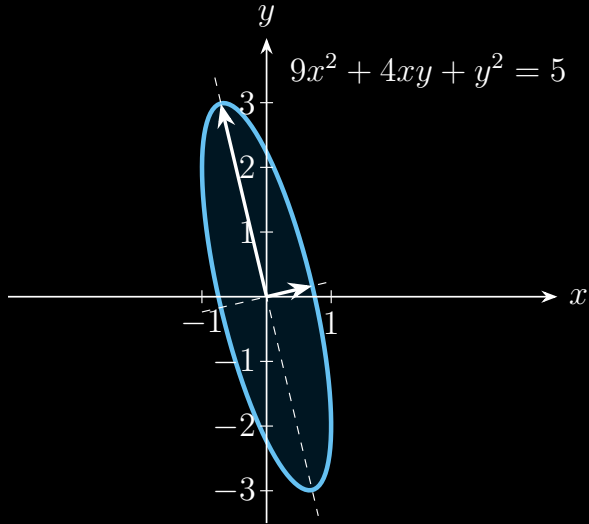
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$



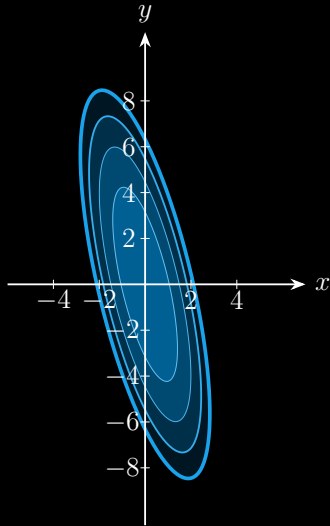
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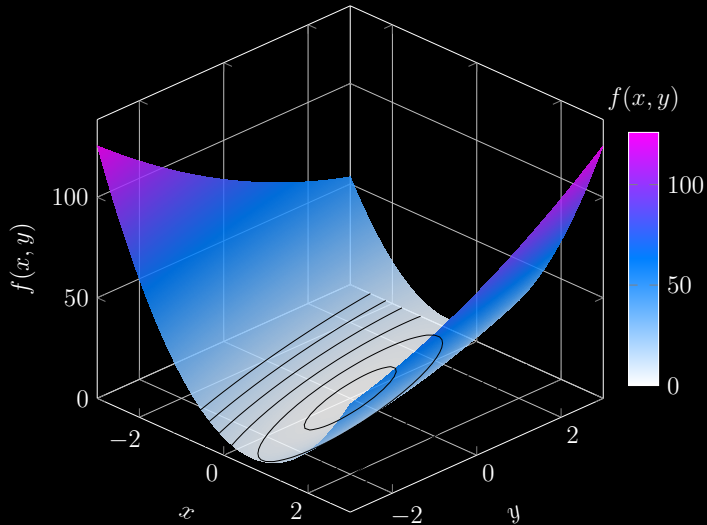
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{w})^2}{(\sqrt{\lambda_2})^2} + \frac{(\mathbf{u}_2^\top \mathbf{w})^2}{(\sqrt{\lambda_1})^2} = 1$



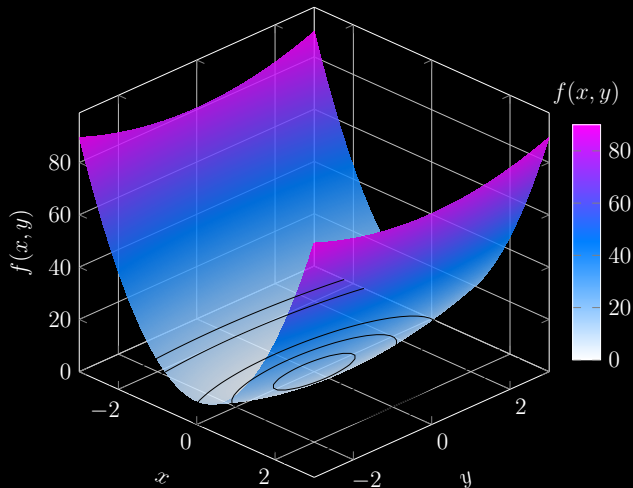
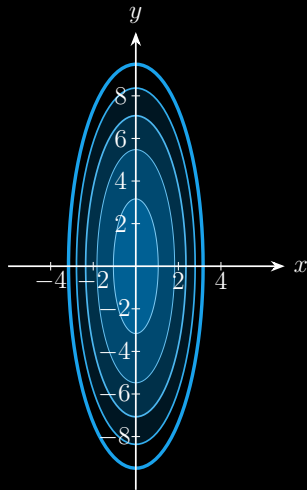
Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



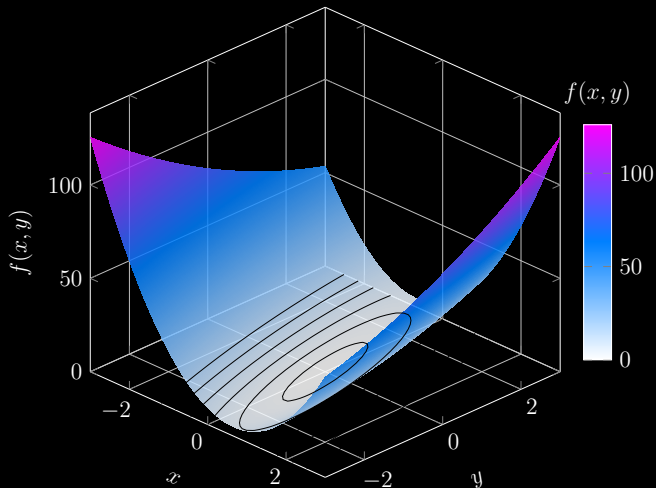
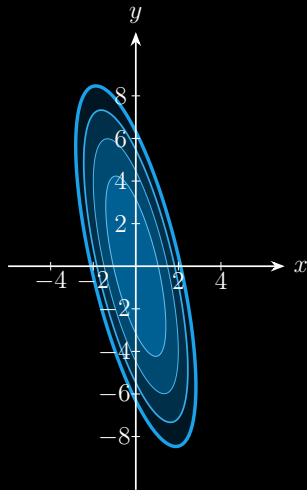
Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



The Geometry of Quadratic Function $f(x, y) = 9x^2 + y^2$



The Geometry of Quadratic Function $f(x, y) = 9x^2 + 4xy + y^2$



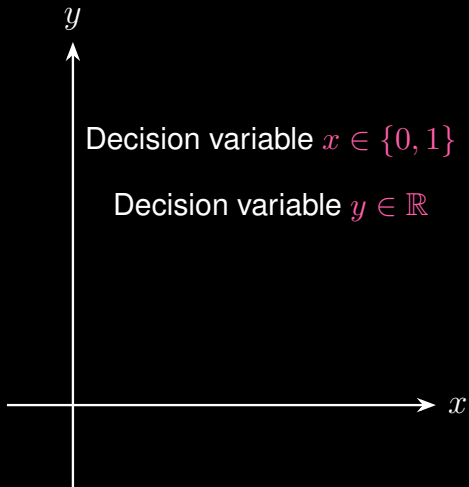
Outline:

- ✓ Linear programming
- ✓ Quadratic programming
- ✓ Geometry of quadratic function
- ✗ Mixed-integer quadratic programming

Mixed-Integer Quadratic Programming

Optimization problem:

$$\begin{aligned} \max_{x \in \{0,1\}, y \in \mathbb{R}} \quad & 9x^2 + y^2 \\ \text{s.t.} \quad & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3 \end{aligned}$$

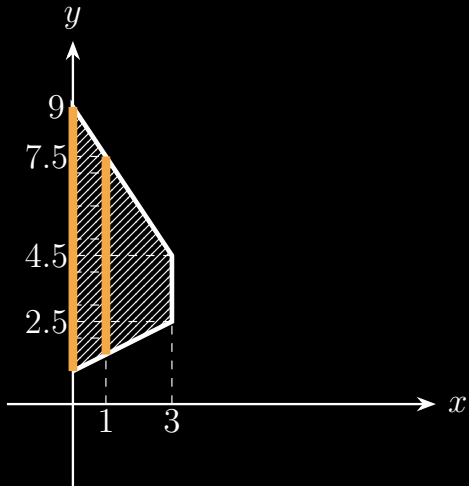


Mixed-Integer Quadratic Programming

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Feasible solutions (—)

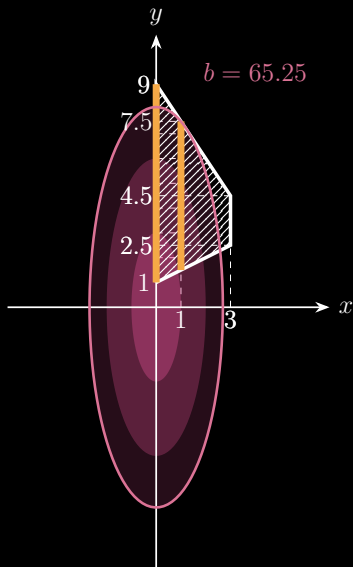


Mixed-Integer Quadratic Programming

Optimization problem:

$$\begin{array}{ll}\max_{x \in \{0,1\}, y \in \mathbb{R}} & 9x^2 + y^2 \\ \text{s.t.} & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3\end{array}$$

Objective $9x^2 + y^2 = b$

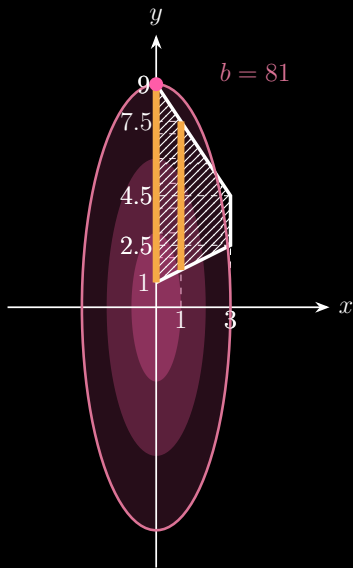


Mixed-Integer Quadratic Programming

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$$\begin{aligned} \max_{x \in \{0,1\}, y \in \mathbb{R}} \quad & 9x^2 + y^2 \\ \text{s.t.} \quad & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3 \end{aligned}$$

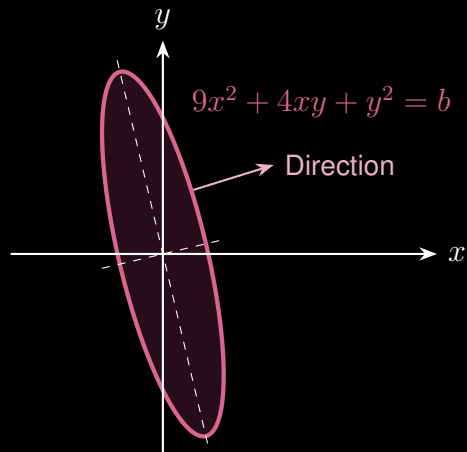
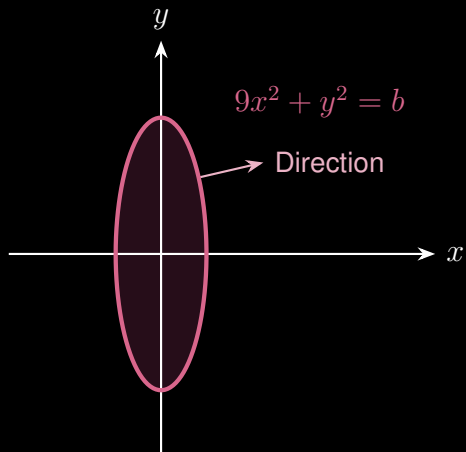
Optimal solution: $x = 0, y = 9$



$$\max_{x,y} 9x^2 + y^2$$

→

$$\max_{x,y} 9x^2 + 4xy + y^2$$



Mixed-Integer Quadratic Programming

Optimization problem:

$$\max_{x \in \{0,1\}, y \in \mathbb{R}} \quad 9x^2 + 4xy + y^2$$

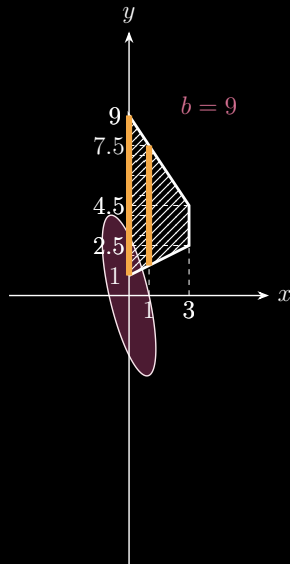
$$\text{s.t.} \quad x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective $9x^2 + 4xy + y^2 = b$



Mixed-Integer Quadratic Programming

Optimization problem:

$$\max_{x \in \{0,1\}, y \in \mathbb{R}} \quad 9x^2 + 4xy + y^2$$

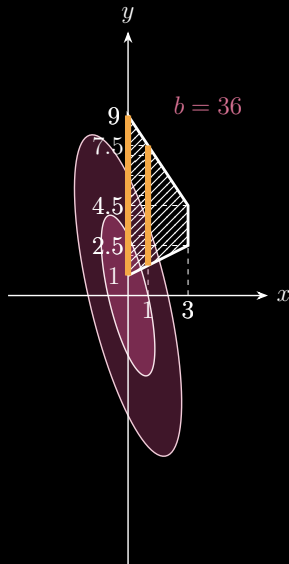
$$\text{s.t.} \quad x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective $9x^2 + 4xy + y^2 = b$



Mixed-Integer Quadratic Programming

Optimization problem:

$$\max_{x \in \{0,1\}, y \in \mathbb{R}} \quad 9x^2 + 4xy + y^2$$

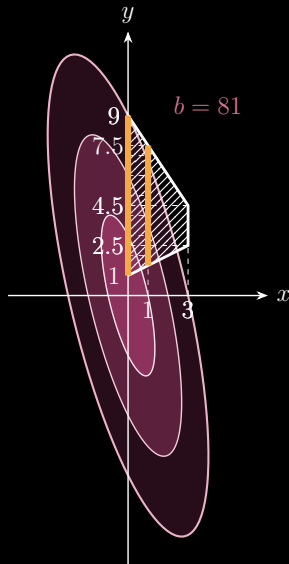
$$\text{s.t.} \quad x, y \geq 0$$

$$3x + 2y \leq 18$$

$$-x + 2y \geq 2$$

$$x \leq 3$$

Objective $9x^2 + 4xy + y^2 = b$

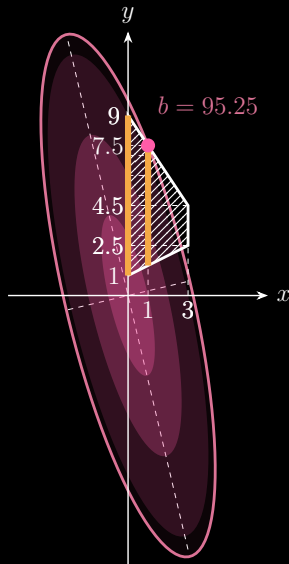


Mixed-Integer Quadratic Programming

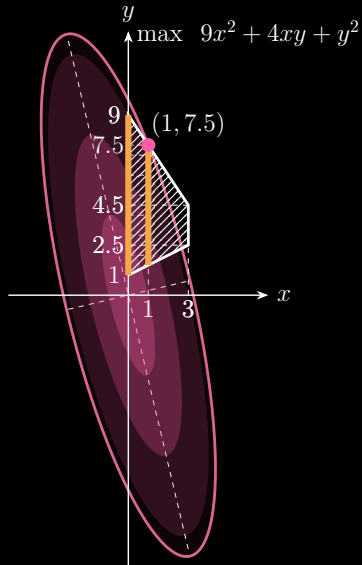
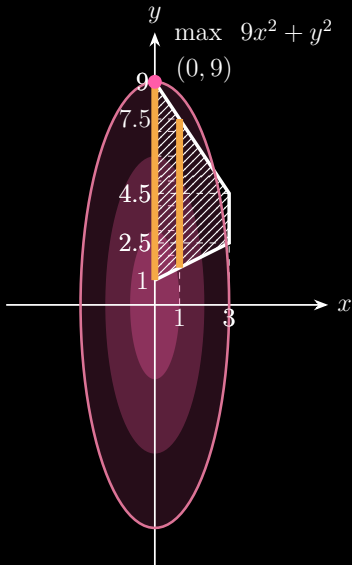
Optimization problem:

$$\begin{aligned} \max_{x \in \{0,1\}, y \in \mathbb{R}} \quad & 9x^2 + 4xy + y^2 \\ \text{s.t.} \quad & x, y \geq 0 \\ & 3x + 2y \leq 18 \\ & -x + 2y \geq 2 \\ & x \leq 3 \end{aligned}$$

Optimal solution: $x = 1, y = 7.5$



Different objectives $\rightarrow$ different optimal solutions



Quick Summary:

- ✓ Linear programming
- ✓ Quadratic programming
- ✓ Geometry of quadratic function
- ✓ Mixed-integer quadratic programming

Thanks for your attention!

About me:

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